

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2012

CL304 Building and Town Planning

Date: 29.11.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Enlist the principles of planning. Differentiate between types of privacy for duplex type bungalows. [05]

(b) Why the economy can be one of the basic points amongst the principles of planning? [02]

Q - 2 (a) Write the abbreviations for the following: [04]

(1)Concrete, (2)Kilogram, (3)Centre to centre, (4)Damp proof coarse.

(b) Define building bye-laws and enlist the bye-laws and regulations for different types of buildings. [05]

(c) Enlist the types of natural growth. Explain any one in detail, with neat sketch. [05]

OR

Q - 2 (a) Write the minimum dimensions of the following: [04]

(1)Weather shed, (2)Slab thickness, (3)Double bed, (4)Kitchen platform.

(b) Write short note on floor area ratio. Explain how is it related to height of the building? Explain the $63\frac{1}{2}^\circ$ rule in brief. [05]

(c) Explain merits and demerits of vertical growth over horizontal growth. [05]

Q - 3 Draw detailed wall thickness plan (**with scale**) for the given **Figure-1**. Also provide the furniture layout and openings wherever it is necessary in the same. Complete with all respect. [14]

OR

Q - 3 Plan a semi-detached type residential building in **Plot No-10**. It should consist of following units. (1) Drawing room, (2) Kitchen (including dining), (3) Master bed room (including Attached toilet), (4) Children's bed room, (5) Bath, (6) W.C., (7) Store room. Draw only Line diagram including compound wall which should fulfill requirements of principles of planning. The area of plot is 300 m^2 and the built-up area is 60%. The details of the plot are given in **Figure-2**. [14]

SECTION – II

- Q - 4 (a) Define slum. Enlist the characteristics of slum. [04]
(b) Explain the principles of town planning, in detail. [03]
- Q - 5 (a) Draw neat sketch of school building and write its components. Enlist the points to be considered while selecting site for school. [07]
(b) Define master plan and write objectives of the master plan. [05]
(c) Give land distribution pattern in a town. [02]

OR

- Q - 5 (a) Draw neat sketch of library. Write principles of planning for library. [07]
(b) Describe various road patterns generally adopted in town planning. Explain merits and demerits of circular pattern. [05]
(c) Define any Two: Shopping centre, Circulation, Zoning. [02]
- Q - 6 (a) Write a note on Indus valley civilization. [06]
(b) Draw perspective view of a stair, consisting of 3 treads and 3 risers. Details of steps are: Riser = 0.2m, Tread = 0.3m, Width of step = 1m. The observer is standing at distance of 2m from one of the corner of step. The height of observer is equals to height of top tread from ground line. [08]

OR

- Q - 6 (a) Explain in detail the planning adopted for Gandhinagar city. [06]
(b) Refer **Figure-3**, and draw its 2-point perspective view, while observing from the distance of 5.0 m from vertical line and height 6.0 m above ground line. [08]

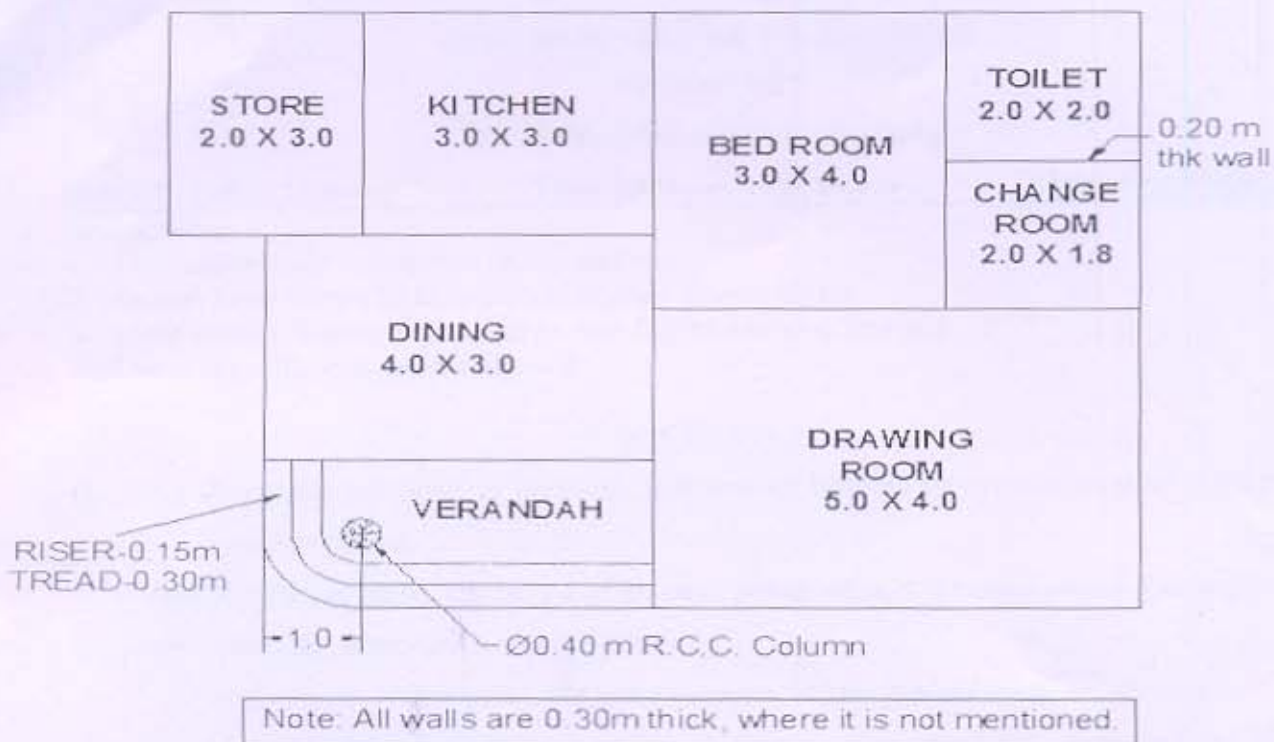


Figure - 1

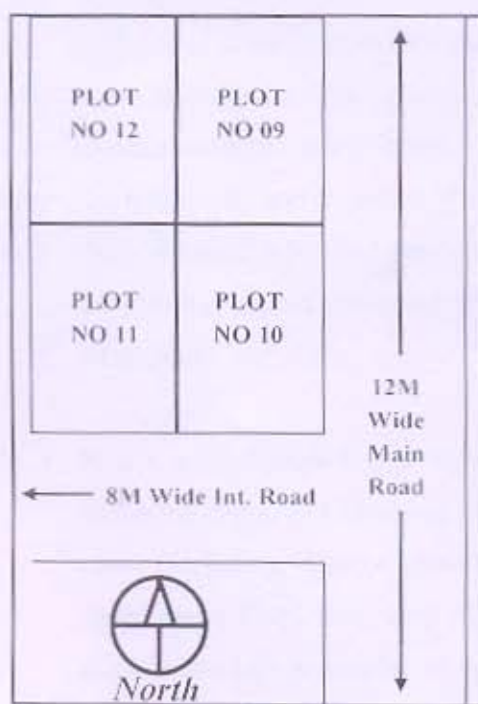


Figure - 2

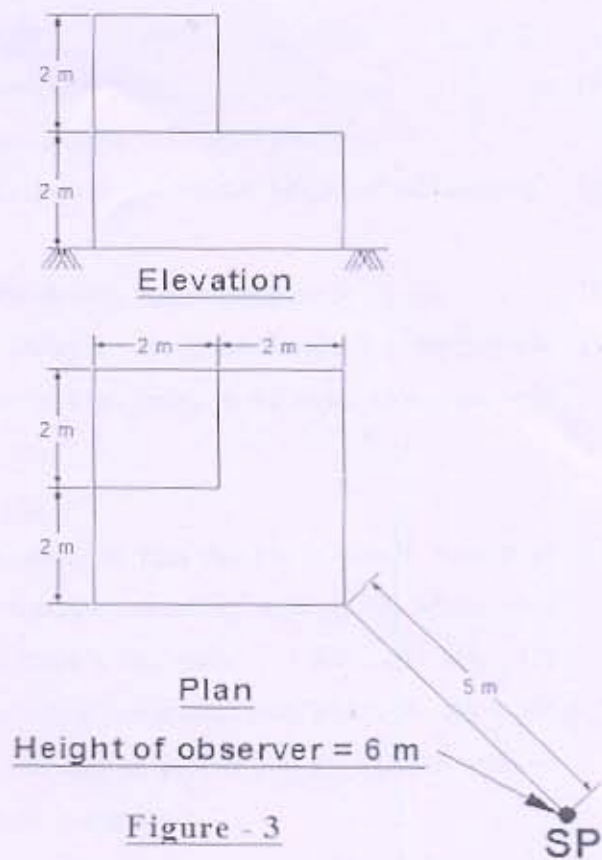


Figure - 3